

Name:

Maths Class:

Year 12
Mathematics Extension 2

HSC Course

Trial

August, 2026

Time allowed: 180 minutes

General Instructions:

- Marks for each question are indicated on the question.
- Approved calculators may be used
- All necessary working should be shown
- Full marks may not be awarded for careless work or illegible writing
- ***Begin each question on a new page***
- Write using black or blue pen
- All answers are to be in the writing booklet provided
- A reference sheet is provided.

Section I Multiple Choice
Questions 1-10
10 Marks

Section II Questions 11-16
90 Marks

Section I

10 marks

Attempt Questions 1-10

Allow about 15 minutes for this section

Use the multiple-choice answer sheet for Questions 1-10

1. What is the scalar product of $19\underset{\sim}{i} - 4\underset{\sim}{j} - 6\underset{\sim}{k}$ and $3\underset{\sim}{i} + 8\underset{\sim}{j} - 7\underset{\sim}{k}$

A. 66

B. 67

C. 76

D. 77

2. $\overrightarrow{OA} = \begin{bmatrix} -2 \\ 1 \\ 1 \end{bmatrix}$ and $\overrightarrow{OB} = \begin{bmatrix} 1 \\ -2 \\ 1 \end{bmatrix}$, which of the following vectors is parallel to $\overrightarrow{AB}$?

A. $\begin{bmatrix} -1 \\ 1 \\ 0 \end{bmatrix}$

B. $\begin{bmatrix} -1 \\ 1 \\ 1 \end{bmatrix}$

C. $\begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}$

D. $\begin{bmatrix} -1 \\ -1 \\ 1 \end{bmatrix}$

3. With a suitable substitution $\int_0^{\frac{\pi}{3}} \tan^5 x \sec^4 x \, dx$ can be expressed as:

A. $\int_1^2 (u^5 + u^7) \, du$

B. $\int_0^{\sqrt{3}} (u^5 + u^7) \, du$

C. $\int_0^{\sqrt{3}} u^3 (u^2 - 1)^2 \, du$

D. $\int_1^2 (u^4 + u^6) \, du$

4. For any two complex numbers z_1 and z_2 which of the following inequalities is always true?

A. $|z_1 - z_2| \geq |z_1| + |z_2|$

B. $|z_1 + z_2| \leq |z_1| - |z_2|$

C. $|z_1 - z_2| \leq ||z_1| - |z_2||$

D. $|z_1 + z_2| \geq ||z_1| - |z_2||$

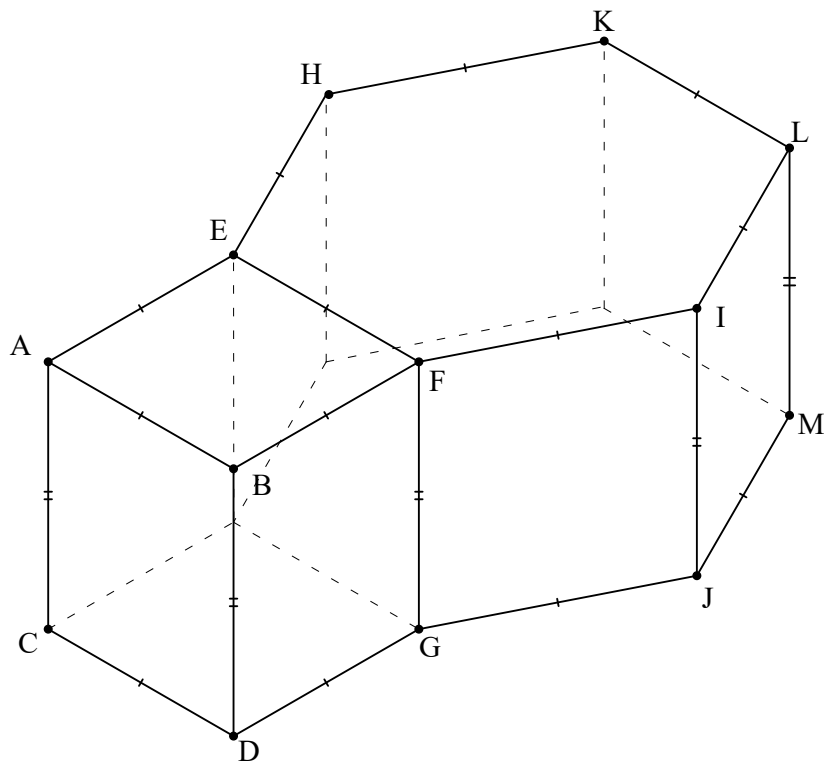
5. $f(x)$ is continuous, differentiable, and integrable over its domain.

$$\int_0^{\frac{\pi}{2}} f(x) \cos x \, dx = 2, \quad f(0) = -1, \quad \text{and} \quad f\left(\frac{\pi}{2}\right) = 3$$

Evaluate $I = \int_0^{\frac{\pi}{2}} f'(x) \sin x \, dx$

- A. $I = 1$
- B. $I = 2$
- C. $I = 5$
- D. $I = 6$
6. Which of the following functions does NOT describe simple harmonic motion
- A. $4 \sin^2 t + 3 \cos 2t$
- B. $3 \sin\left(3t + \frac{\pi}{6}\right) - 2 \cos\left(3t - \frac{\pi}{3}\right)$
- C. $5 \sin 4t + 12 \cos 4t - 3$
- D. $3 \sin 5t - 2 \cos 3t$

7.



A 3D shape is formed by placing two right prisms side by side, as shown in the diagram. One prism has a square base $ABFE$ and the other has a regular hexagonal base $EFILKH$. Both prisms have the same height, and all base edges have the same length.

Which of the following represents $\overrightarrow{DK}$

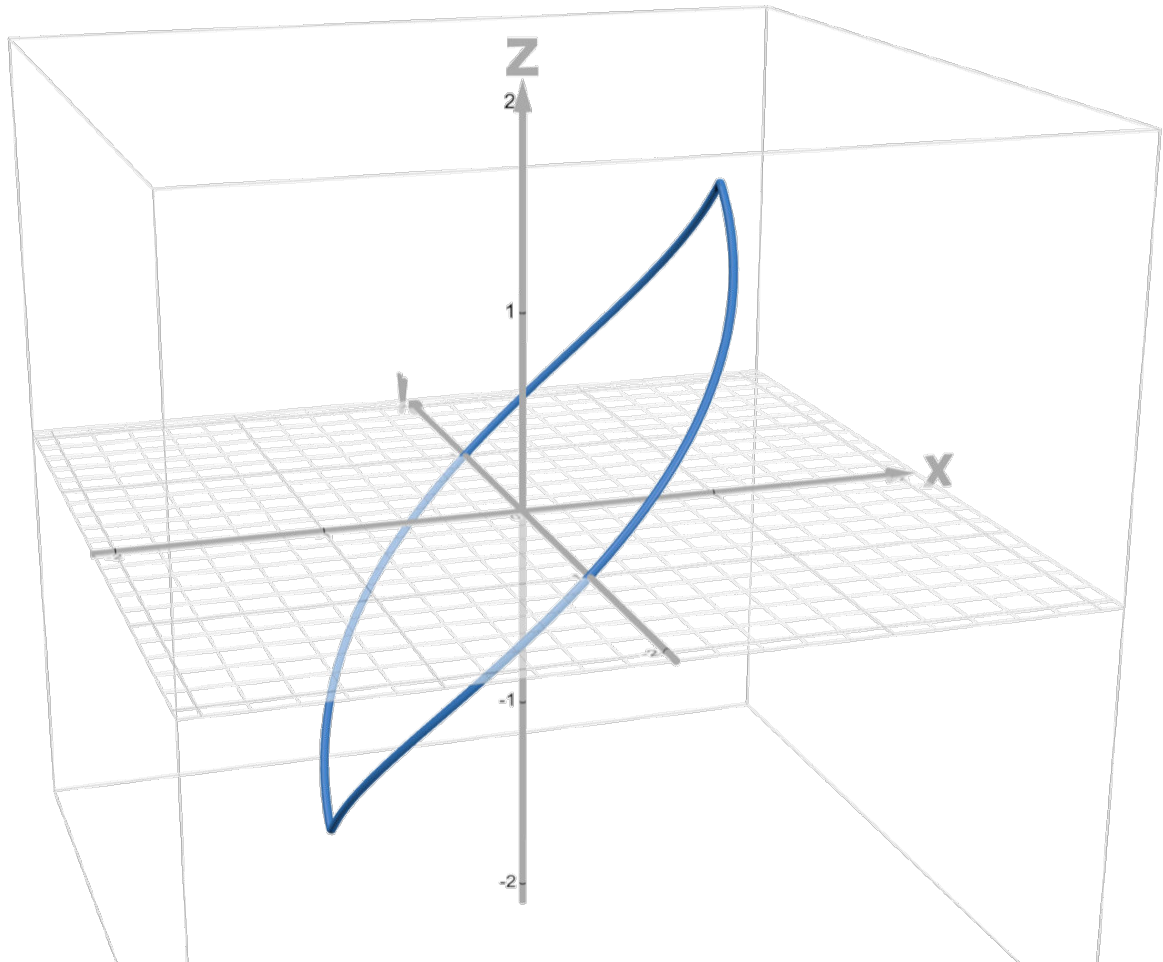
A. $\overrightarrow{DK} = \overrightarrow{AB} - \overrightarrow{AC} + \overrightarrow{AE} - \overrightarrow{HE} + \overrightarrow{HK}$

B. $\overrightarrow{DK} = \overrightarrow{DG} + \overrightarrow{GJ} + \overrightarrow{JL} - \overrightarrow{KL}$

C. $\overrightarrow{DK} = \overrightarrow{AE} + \overrightarrow{HK} - \overrightarrow{AB} - \overrightarrow{FG} + \overrightarrow{IL}$

D. $\overrightarrow{DK} = \overrightarrow{AE} + \overrightarrow{HK} + \overrightarrow{AB} - \overrightarrow{AC} + \overrightarrow{IL}$

8.



Which of the following best represents the parametric equation of the 3D curve shown above?

- A. $(\cos \pi t, \sin \pi t, \arcsin(\sin \pi t)), -2\pi \leq t \leq 2\pi$
- B. $(\sin \pi t, \cos \pi t, \arcsin(\sin \pi t)), -2\pi \leq t \leq 2\pi$
- C. $(\sin \pi t, \cos \pi t, \arccos(\cos \pi t)), -2\pi \leq t \leq 2\pi$
- D. $(\cos \pi t, \arccos(\cos \pi t), \sin \pi t), -2\pi \leq t \leq 2\pi$

9. The intersection of four regions on the Argand diagram forms a kite with diagonals running along the real and imaginary axes. The regions are defined by:

$$\arg(z - i) \leq -\frac{\pi}{3}$$

$$\arg(z + i) \leq \frac{\pi}{3}$$

$$\arg(z + \sqrt{3}) \leq \frac{\pi}{6}$$

$$\arg(z + \sqrt{3}) \geq -\frac{\pi}{6}$$

What is the exact area of the kite?

A. $\frac{2}{\sqrt{3}}$

B. $\frac{4}{\sqrt{3}}$

C. $\frac{8}{\sqrt{3}}$

D. $2\sqrt{3}$

10. $f(x)$ is a continuous and integrable function over the real numbers.

Given that $f(2x) = 3f(x)$ and $\int_0^1 f(x) dx = 2$

find $\int_1^2 f(x) dx$

A. 4

B. 6

C. 10

D. 12

Please turn over for Section II

Section II

Attempt Questions 11-16 (90 marks)

Allow about 2 hours and 45 minutes for this section

Answer each question on a new page in the answer booklet

For questions in Section II, your responses **must include** relevant mathematical reasoning and/or calculations.

Question 11 (15 marks) Start a new page	Marks
a) Find $\int 2\cos 5x \cos 4x \, dx$	2
b) i) Write the number $1 - \sqrt{3}i$ in exponential form	2
ii) Hence, or otherwise write $(1 - \sqrt{3}i)^6$ in exact cartesian form	2
c) Find the values of α , and β such that the points $(-3, \alpha, \beta)$, $(5, 4, 2)$, and $(3, 12, 2)$ are collinear	2
d) Solve the quadratic equation $z^2 - 6z + 13 = 0$ over the complex numbers giving your answer in Cartesian form	2
e) O and A are two points such that $\overrightarrow{OA} = \underline{\underline{i}} + \underline{\underline{j}} - 4\underline{\underline{k}}$ and $\overrightarrow{OB} = 5\underline{\underline{i}} + 4\underline{\underline{j}} + 3\underline{\underline{k}}$	
i) Find the angle between $\overrightarrow{OA}$ and $\overrightarrow{OB}$ to the nearest degree	3
ii) Find the exact area of triangle OAB	2

End of Question 11

a) Evaluate $\int_1^e \frac{(\ln x)^3}{x} dx$ 2

b) Solve $z = \sqrt[3]{i}$, giving your answers in cartesian form 3

c) Shade the region in the complex plane defined by 3

$$1 < \operatorname{Im}(z) \leq 4 \quad \text{and} \quad \frac{\pi}{4} \leq \arg(z) \leq \frac{3\pi}{4}$$

d) A particle is moving in simple harmonic motion described by $\ddot{x} = -9(x - 2)$ 3

When the particle passes through the point $x = -1$ m its speed is $6\sqrt{3} \text{ ms}^{-1}$

Find the distance the particle travels during a full period of its motion

e) The points $A = (-1, 0, 0)$, $B = (0, 1, 0)$, and $C = (0, 0, 1)$ are 3 of the vertices of a regular unit octahedron centred on the origin

i) Find $\operatorname{proj}_{\overrightarrow{BC}} \overrightarrow{AB}$ 2

ii) Hence or otherwise find the shortest distance from A to the line BC 2

End of Question 12 please turn over

Question 13 (15 marks) Start a new page**Marks**

a) Evaluate $\sum_{n=1}^{2026} i^n$ 2

b) i) Show that $x + \frac{1}{x} \geq 2$ for all positive real numbers x 1

ii) Prove that for four positive real numbers $a, b, c, d, a \neq b \neq c \neq d$ 2

$$\left(\frac{a}{b} - \frac{c}{d}\right)\left(\frac{b}{a} - \frac{d}{c}\right) \leq 0$$

c) Use the substitution $u^2 = 5 - x, u > 0$ to show that 4

$$\int_1^4 \frac{3}{x + 4\sqrt{5-x}} dx = \ln\left(\frac{2048}{729}\right)$$

d) Consider the complex number $z = \cos \theta - i \sin \theta$

i) Use De Moivre's theorem to show that $z^n + \frac{1}{z^n} = 2 \cos n\theta, n \in \mathbb{Z}$ 2

ii) Hence or otherwise express $\left(z + \frac{1}{z}\right)^5$ in the form: 2

$$A \cos(5\theta) + B \cos(3\theta) + C \cos(\theta) \text{ where } A, B, C \in \mathbb{R}$$

iii) Hence or otherwise, evaluate $\int_0^{\frac{\pi}{2}} \cos^5 \theta d\theta$ 2

End of Question

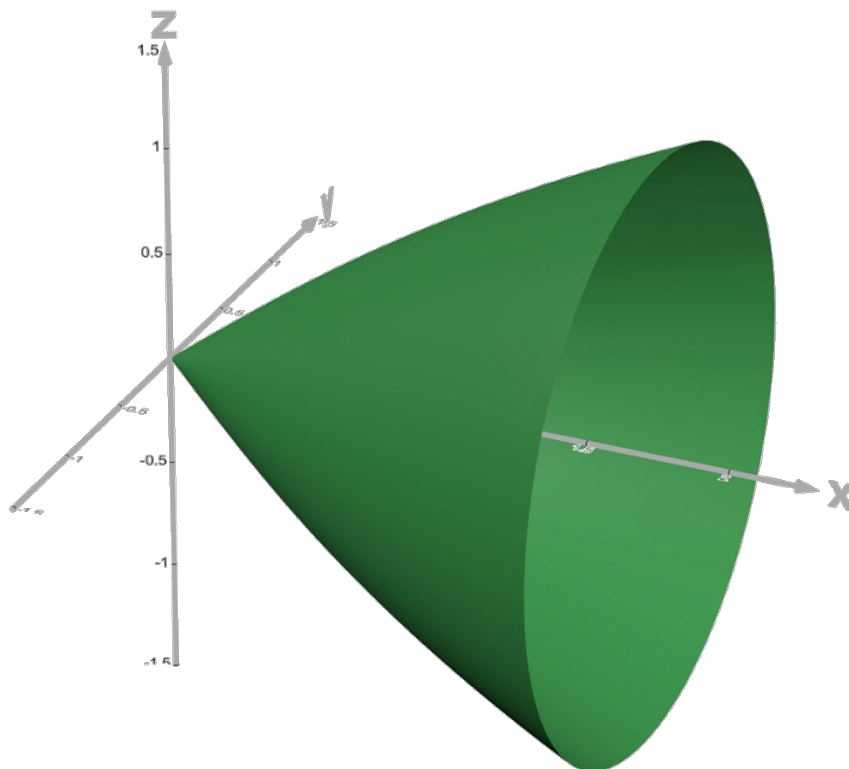
a) A particle initially at the origin moves in a straight line with an initial velocity of

1 m/s such that its acceleration $a \text{ m/s}^2$ is given by $a = v \sqrt{4 - v^2}$

i) Find its exact velocity when its displacement $x = \frac{\pi}{6}$ 3

ii) With relevant mathematical reasoning, describe the eventual motion of the particle 1

b)



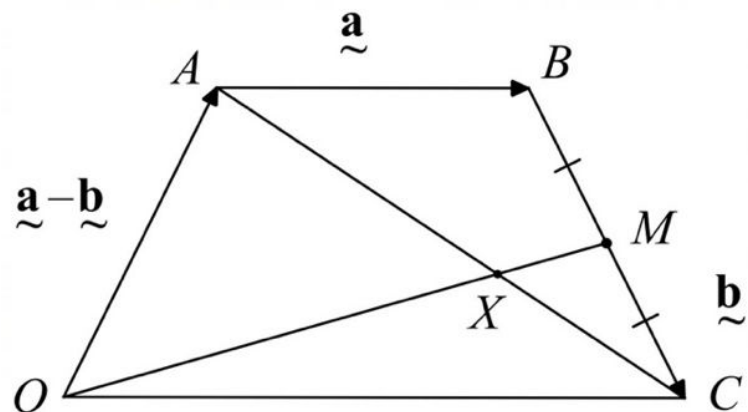
Find the exact volume of the solid formed when $y = \sqrt{2x} \arctan x$ is rotated about the x – axis, between $x = 0$ and $x = \sqrt{3}$ 3

Question 14 continues over the page

c) Prove by contradiction that $4x^2 - y^2 = 1$ does not have any integer solutions

3

d) $OABC$ is a quadrilateral



$$\overrightarrow{AB} = \underset{\sim}{a}, \overrightarrow{BC} = \underset{\sim}{b}, \text{ and } \overrightarrow{OA} = \underset{\sim}{a} - \underset{\sim}{b}$$

M is the midpoint of BC

X is the point on OM such that $OX:XM = n : 1$

X also lies on the line AC

i) Express $\overrightarrow{OC}$ in terms of $\underset{\sim}{a}$ and/or $\underset{\sim}{b}$ in its simplest form

1

ii) Express $\overrightarrow{OX}$ in terms of $\underset{\sim}{a}$ and $\underset{\sim}{b}$ and hence show that

2

$$\overrightarrow{AX} = \frac{n-1}{n+1} \underset{\sim}{a} + \frac{n+2}{2(n+1)} \underset{\sim}{b}$$

iii) Find the value of n

2

End of Question

a) What is the contrapositive of the following implication

1

$$\neg(A \Rightarrow C) \Rightarrow (\neg(A \Rightarrow B) \text{ OR } \neg(B \Rightarrow C))$$

b) Let $I_n = \int_0^1 x(1-x)^n dx$ where n is positive integer

i) Show that $I_n = \frac{1}{(n+1)(n+2)}$

3

ii) Hence by considering the partial fraction decomposition of I_n , find a closed form algebraic expression for S_n , the sum of the first n terms of I_n

2

$$S_n = \sum_{k=1}^n I_k$$

c) Use mathematical induction to prove that, for all positive integers n

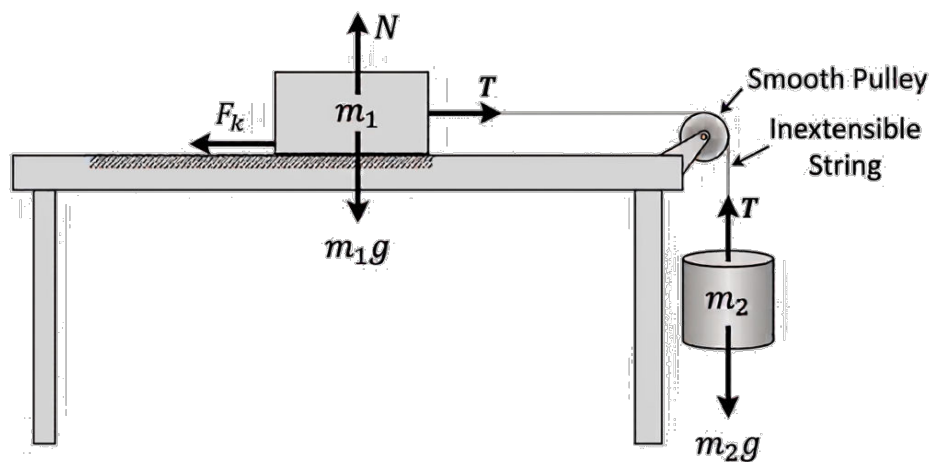
4

$$\tan \theta + 2 \tan 2\theta + \dots + 2^{n-1} \tan(2^{n-1}\theta) = \cot \theta - 2^n \cot(2^n \theta)$$

d) Use the fundamental theorem of calculus to show that $\frac{d}{dx} \left[\int_1^{x^2} \ln t \, dt \right] = 4x \ln x$ 2

e) A light, inextensible string passes over a smooth, fixed pulley. 3

One end is attached to a mass $m_1 = 8 \text{ kg}$ on a rough horizontal table, where a kinetic friction force $F_k = u_k N$ newtons acts in the opposite direction to that of the motion, where u_k is the coefficient of kinetic friction and N is the normal force acting on m_1 . The other end of the string hangs vertically and is attached to a mass $m_2 = 6 \text{ kg}$



The coefficient of kinetic friction $u_k = \frac{1}{4}$

The system is released from rest

In terms of gravity g , what is the tension T in the string ?

End of Question 15

- a) i) Prove that the product of two consecutive positive even integers is divisible by 8 2
 ii) Hence prove that for all prime numbers $P > 3$, $P^2 - 1$ is divisible by 24 3

- b) An exploratory lander on the distant exoplanet Arrakis uses a high-powered magnetic launcher to fire a sampling probe of mass m kg vertically upwards. The launcher is positioned at $x = 1$, where x represents the displacement from the surface in meters.

Driven by magnetic propulsion from the launcher, the probe moves through a dense plasma atmosphere as it travels upwards under a gravity of g m/s²

The combined net non-gravitational force F_r acting on the metal probe is modelled by

$$F_r = \frac{mg}{x^3} (1 + v^2) - mgv^2$$

Where v is the speed of the probe

The lander's booster launches the probe with an initial upward speed of U m/s

- i) Show that the equation of motion while the probe is moving upwards can be written as: 1

$$\frac{v}{1 + v^2} \frac{dv}{dx} = -g \left(1 - \frac{1}{x^3} \right)$$

- ii) Hence, show that the greatest height of the probe $x = H$, satisfies: 3

$$\ln (1 + U^2) = 2g(H - 1) - g \left(\frac{H^2 - 1}{H^2} \right)$$

Question 16 continues over the page

c) A and B are two distinct points in three-dimensional space such that

$$\overrightarrow{OA} = \begin{pmatrix} 6 \\ 4 \\ -2 \end{pmatrix} \text{ and } \overrightarrow{OB} = \begin{pmatrix} 4 \\ 8 \\ 2 \end{pmatrix}$$

i) Let S_1 be the set of all points P such that $\overrightarrow{AP} \cdot \overrightarrow{BP} = 0$.

2

Find the vector equation of S_1

ii) let M be the midpoint of the line AB and S_2 be the set of all points N such that

1

$$|\overrightarrow{AN}| = |\overrightarrow{MN}|$$

The intersection of S_1 and S_2 is the circle S . Find the radius of S

d) i) Show that $\int_a^b f(x)dx = \int_a^b f(a+b-x)dx$

1

ii) Hence or otherwise find the exact value of $\int_{\frac{\pi}{6}}^{\frac{\pi}{3}} \frac{1}{1 + \sqrt{\tan x}} dx$

2

2026 EX2 Trial solutions

1 2 3 4 5 6 7 8 9 10
 (B) (A) (B) (D) (A) (D) (C) (B) (B) (C)

1) $19 \times 3 - 4 \times 8 + 6 \times 7 = 67 \therefore (B)$

2) $AB = \begin{bmatrix} 1 & -2 \\ -2 & -1 \\ 1 & -1 \end{bmatrix} = \begin{bmatrix} 3 \\ 3 \\ 0 \end{bmatrix} = -\frac{1}{3} \begin{bmatrix} -1 \\ -1 \\ 0 \end{bmatrix} \therefore (A)$

3)
$$I = \int_0^{\pi/3} \tan^5 x \sec^4 x \, dx$$

let $u = \tan x$
 $du = \sec^2 x \, dx$
 $dx = \frac{du}{\sec^2 x}$

$x=0 \quad u=0$
 $x=\pi/3 \quad u=\sqrt{3}$

$$= \int_{u=0}^{\sqrt{3}} u^5 \cdot (1 + \tan^2 x) \frac{\sec^2 x \, du}{\sec^2 x}$$

$$= \int_0^{\sqrt{3}} u^5 (1 + u^2) \, du \therefore (B)$$

4) A false as $|z_1 - z_2| < |z_1| + |z_2|$ e.g. if signs same i.e. $z_1=1 \quad z_2=-1$
 B false again if signs same.

e.g. $|z_1 + z_2| > |z_1| - |z_2|$ e.g. $z_1=1 \quad z_2=1$

C false if signs different e.g. $z_1=1 \quad z_2=-1$

$|z_1 - z_2| > ||z_1| - |z_2||$

(D) correct answer This is in fact. The new triangle inequality.

5) $\int_0^{\pi/2} f(x) \cos x \, dx = 2$ $f(\pi/2) = 3$ let $u = f(x)$ $v' = \cos x$
 $f(0) = -1$ $\frac{du}{dx} = f'(x)$ $v = \sin x$

$$uv \Big|_0^{\pi/2} - \int_0^{\pi/2} v \frac{du}{dx} dx = 2$$

$$\left[f(x) \sin x \right]_0^{\pi/2} - \int_0^{\pi/2} f'(x) \sin x \, dx = 2$$

$$f(\pi/2) \times \sin \pi/2 - f(0) \times \sin 0 - I = 2$$

$$3 \times 1 - -1 \times 0 - 2 = I$$

$$\therefore I = 3 - 2 = 1 \quad \therefore \textcircled{A}$$

6) $\textcircled{D}$ as in A $\sin^2 t$ can be expressed in terms of $\cos 2t$, in B and C can use auxiliary angle to express as one trig function (C has an offset). However in D periods different so although periodic, it is not SHM.

7) need 5 correct unique vectors to go from D to K
eliminate B as only 4

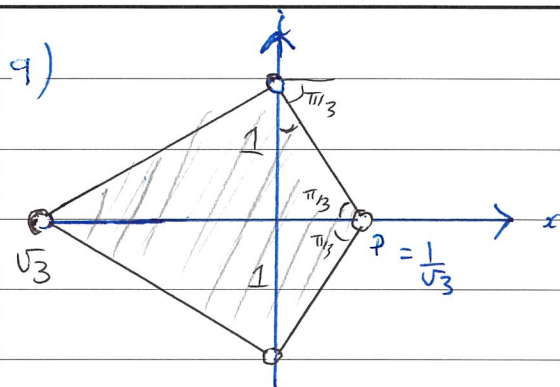
we note $\vec{AB}$ wrong direction need $\vec{BA}$ or $-\vec{AB}$ so
we can eliminate A and D $\therefore \textcircled{C}$

checking $\vec{AE} = \vec{DG}$ $-\vec{FG} = \vec{GF}$ $-\vec{AB} = \vec{FE}$ $\vec{IL} = \vec{FH}$ and $\vec{HI} \checkmark$

8) $\arccos$ has range $[0, \pi]$ $\therefore$ eliminate C and D.

$\begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}$ on curve so $x=0$ and $z=0$ at same time.

$\sin t = 0$ when $t=0$ or π or $\sin(\sin \pi t)$ also $= 0$ when $t=0, \pi$
 $\therefore \textcircled{B}$



$$\tan \pi/3 = \frac{1}{p}$$

$$\sqrt{3} = \frac{1}{p} \quad \therefore p = \frac{1}{\sqrt{3}}$$

$$A = \frac{1}{2}(a+b) = \frac{1}{2} \times (1+1) \times \left(\frac{1}{\sqrt{3}} + \sqrt{3}\right)$$

note: after area, The holes at intersection do not matter.

$$= \frac{\sqrt{3} \times \sqrt{3} + 1}{\sqrt{3}}$$

$$= \frac{4}{\sqrt{3}} \quad \therefore \textcircled{B}$$

10) $f(2x) = 3f(x)$

$$\therefore f(x) = \frac{f(2x)}{3}$$

$$\int_0^1 f(x) dx = 2 \quad (\text{given})$$

$$\int_0^1 \frac{f(2x)}{3} dx = 2$$

$$\int_{u=0}^{u=2} \frac{f(u)}{3} \times \frac{du}{2} = 2$$

$$\therefore \int_0^2 f(u) du = 2 \times 6 \quad \text{but } u \text{ a dummy variable.} \quad \therefore \int_0^2 f(x) dx = 12$$

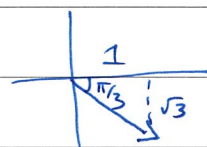
$$\text{and } \int_1^2 f(x) dx = \int_0^2 f(x) dx - \int_0^1 f(x) dx$$

$$= 12 - 2 = 10 \quad \therefore \textcircled{C}$$

— end MC —

$$\begin{aligned}
 \text{Q11)} \quad (a) \quad \int 2 \cos 5x \cos 4x dx &= 2 \cdot \frac{1}{2} \int (\cos(5x+4x) - \sin(5x-4x)) dx \\
 &= \int \cos 9x dx - \int \sin x dx \\
 &= \frac{\sin 9x}{9} + \cos x + C
 \end{aligned}$$

$$(b) (i) \quad 1 - \sqrt{3}i = 2e^{-i\pi/3}$$



$$\begin{aligned}
 (ii) \quad (1 - \sqrt{3}i)^6 &= (2e^{-i\pi/3})^6 \\
 &= 2^6 \cdot e^{-6i\pi/3} = 64e^{-2\pi i} = 64e^{2\pi i} \\
 &= 64
 \end{aligned}$$

(c) to be colinear all on same line.

so for example $(-3, \alpha, \beta)$ on line through $(5, 4, 2)$ and $(3, 13, 2)$

$$i.e. \quad \begin{pmatrix} -3 \\ \alpha \\ \beta \end{pmatrix} = \begin{pmatrix} 5 \\ 4 \\ 2 \end{pmatrix} + \lambda \begin{pmatrix} 5-3 \\ 4-12 \\ 2-2 \end{pmatrix}$$

$$\begin{pmatrix} -3 \\ \alpha \\ \beta \end{pmatrix} = \begin{pmatrix} 5 \\ 4 \\ 2 \end{pmatrix} + \lambda \begin{pmatrix} 2 \\ -8 \\ 0 \end{pmatrix}$$

$$\text{so } \beta = 2 \quad \text{and} \quad 2\lambda + 5 = -3 \quad \therefore \lambda = -4$$

from z from x

$$\text{from y. } \alpha = 4 + -8\lambda = 4 + 32 = 36.$$

$$i.e. \quad \alpha = 36 \text{ and } \beta = 2$$

Q11) d)

$$z^2 - 6z + 13 = 0$$

$$a=1 \quad b=-6 \quad c=13$$

$$z = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

$$= \frac{-(-6) \pm \sqrt{(-6)^2 - 4 \times 1 \times 13}}{2} = \frac{6 \pm \sqrt{36 - 52}}{2}$$

$$= \frac{6 \pm \sqrt{-16}}{2}$$

$$z = 3 + 2i \quad \text{or} \quad 3 - 2i$$

(e) $\vec{OA} \cdot \vec{OB} = |\vec{OA}| |\vec{OB}| \cos \theta$

$$\begin{pmatrix} 1 \\ 1 \\ -4 \end{pmatrix} \cdot \begin{pmatrix} 5 \\ 4 \\ 3 \end{pmatrix} = \sqrt{1^2 + 1^2 + (-4)^2} \cdot \sqrt{5^2 + 4^2 + 3^2} \cos \theta$$

$$\sqrt{18} \cdot \sqrt{50} \cos \theta = 5 + 4 - 12$$

$$\cos \theta = \frac{-3}{\sqrt{900}} = -\frac{3}{30}$$

So $|\vec{OA}| |\vec{OB}| = 30$

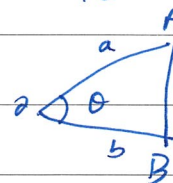
$$\cos \theta = -\frac{1}{10} \quad \therefore \theta = 96^\circ$$

(ii) in Q2 so $\sin \theta$ +ve.

$$\begin{array}{c} 10 \\ \swarrow \\ 10 \end{array} \quad \begin{array}{c} \searrow \\ \sqrt{99} \\ \downarrow \\ 1 \end{array} \quad \therefore \sin \theta = \frac{\sqrt{99}}{10} = \frac{3\sqrt{11}}{10}$$

$$A = \frac{1}{2} ab \sin \theta = \frac{1}{2} \cdot 30 \cdot \frac{3\sqrt{11}}{10}$$

$$A = \frac{9\sqrt{11}}{2} \text{ u}^2$$



(END Q11)

Question 12

$$(a) \int_1^e \frac{(\ln x)^3}{x} dx = \int_0^1 u^3 du$$

$$= \left[\frac{u^4}{4} \right]_0^1 = \underline{\underline{\frac{1}{4}}}$$

$$u = \ln x$$

$$\frac{du}{dx} = \frac{1}{x}$$

$$\cdot du = \frac{dx}{x}$$

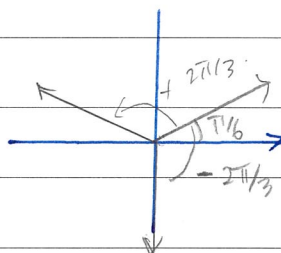
$$x=1 \quad u=0$$

$$x=e \quad u=1$$

$$(b) i = \text{cis}(\pi/2)$$

$$z^3 = \text{cis}(\pi/2)$$

$$\therefore z = \text{cis}(\pi/2 \div 3)$$

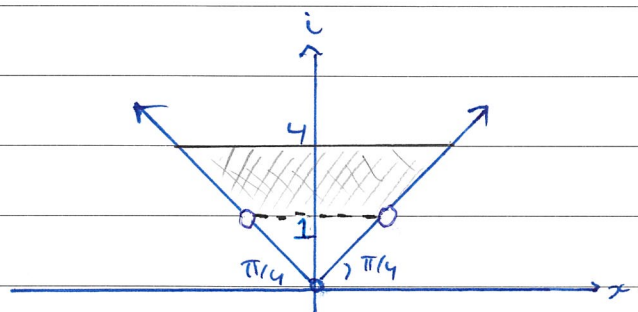


$$z = \text{cis}(\pi/6), \text{cis}(\pi/6 + 2\pi/3), \text{cis}(\pi/6 - 2\pi/3)$$

$$= \text{cis} \pi/6, \text{cis} 5\pi/6, \text{cis}(-\pi/2)$$

$$= \sqrt{3} + \frac{1}{\sqrt{3}}i, -\sqrt{3} + \frac{1}{\sqrt{3}}i, -i$$

(c)



$$(d) x'' = -9(x-2)$$

$$= -3^2(x-2)$$

SHM

$$n=3$$

$$c=2$$

(p12d)

Cont....

So

$$x = a \sin(\pi t + \alpha) + c$$

$$x = a \sin(3t + \alpha) + 2$$

$$\dot{x} = 3a \cos(3t + \alpha)$$

let $\beta = 3t + \alpha$
(we don't need α , clean)

note we are not told initially, and are only after amplitude, not interested in position at certain time, so the phase and α not needed.

$$x = a \sin(\beta) + 2$$

$$\dot{x} = 3a \cos(\beta)$$

at the same time or β $x = -1$ $\dot{x} = 6\sqrt{3}$

So $-1 = a \sin(\beta) + 2 \Rightarrow a \sin(\beta) = -3$

$6\sqrt{3} = 3a \cos(\beta) \Rightarrow a \cos(\beta) = 2\sqrt{3}$

pick the
so we square
anyway and
only after
amplitude.

$$(a \sin(\beta))^2 + (a \cos(\beta))^2 = (-3)^2 + (2\sqrt{3})^2$$

$$a^2 (\sin^2 \beta + \cos^2 \beta) = 21$$

$$\therefore a = \sqrt{21} \text{ m}$$

for full Period $\frac{a}{\omega} \cdot \frac{1}{a} \cdot \frac{a}{\omega} \cdot \frac{a}{\omega}$ distance travelled = $4a$

$\therefore$ distance travelled in full period = $4\sqrt{21}$

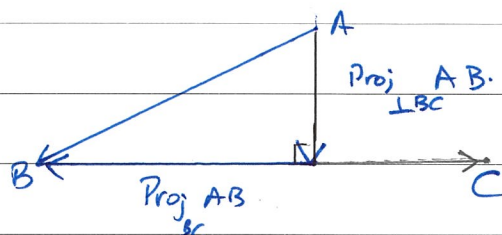
e) $\vec{OA} = \begin{pmatrix} -1 \\ 0 \\ 0 \end{pmatrix}$ $\vec{OB} = \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}$ $\vec{OC} = \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}$

$\vec{AB} = \vec{OB} - \vec{OA} = \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix}$ $\vec{BC} = \vec{OC} - \vec{OB} = \begin{pmatrix} 0 \\ -1 \\ 1 \end{pmatrix}$

Proj $\vec{AB}$ on $\vec{BC} = \frac{\vec{AB} \cdot \vec{BC}}{|\vec{BC}|^2} \times \vec{BC} = \frac{1 \times 0 + 1 \times -1 + 0 \times 1}{(\sqrt{2})^2} \begin{pmatrix} 0 \\ -1 \\ 1 \end{pmatrix} = \begin{pmatrix} 0 \\ -1/2 \\ 1/2 \end{pmatrix}$

Q12(e) cont

(ii)



The shortest line from A to line BC is

The magnitude of Projection of AB onto the Perpendicular line of BC ($\text{Proj}_{AD} AB$)

$$\text{Proj}_{AD} AB = AB - \text{Proj}_{BC} AB$$

$$= \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix} - \begin{pmatrix} 0 \\ \frac{1}{2} \\ -\frac{1}{2} \end{pmatrix} = \begin{pmatrix} 1 \\ \frac{1}{2} \\ \frac{1}{2} \end{pmatrix}$$

$$d_{\min} = |\text{Proj}_{AD} AB| = \sqrt{1^2 + \frac{1}{2}^2 + \frac{1}{2}^2} = \sqrt{\frac{6}{4}}$$

$$\underline{d_{\min} = \frac{\sqrt{6}}{2}}$$

Question 13

$$\begin{aligned} \text{(a)} \quad \sum_{i=1}^{2026} i^n &= i + i^2 + i^3 + i^4 + i^5 + \dots + i^{2024} + i^{2025} + i^{2026} \\ &= \underbrace{i - 1 - i + 1 + i + \dots + 1 + i - 1}_{=0} \underbrace{\dots}_{=0 \text{ as divisible by 4}} \\ &= i - 1 \end{aligned}$$

Q13 continued

$$(b) (i) \left(\sqrt{x} - \frac{1}{\sqrt{x}}\right)^2 \geq 0$$

$$(\sqrt{x})^2 + \left(-\frac{1}{\sqrt{x}}\right)^2 - \frac{2\sqrt{x}}{\sqrt{x}} \geq 0$$

$$x + \frac{1}{x} - 2 \geq 0$$

$$\therefore x + \frac{1}{x} \geq 2 \text{ as required.}$$

$$(ii) \text{ LHS} = \left(\frac{a}{b} - \frac{c}{d}\right)\left(\frac{b}{a} - \frac{d}{c}\right)$$

$$= \frac{a \cdot b}{b \cdot a} - \frac{c \cdot b}{d \cdot a} - \frac{a \cdot d}{b \cdot c} + \frac{c \cdot d}{d \cdot c}$$

$$= 1 - \frac{bc}{da} - \frac{ad}{bc} + 1$$

$$= 2 - \left(\frac{bc}{ad} + \frac{ad}{bc}\right)$$

$$\text{from Part (i) let } x = \frac{bc}{ad}$$

$$\leq 2 - 2$$

$$x + \frac{1}{x} \geq 2$$

$$= 0$$

$$-(x + \frac{1}{x}) \leq -2$$

$$\therefore \left(\frac{a}{b} - \frac{c}{d}\right)\left(\frac{b}{a} - \frac{d}{c}\right) \leq 0 \text{ as required.}$$

Q13) b(ii) alternate solution

$$LHS = \left(\frac{a}{b} - \frac{c}{d} \right) \left(\frac{b}{a} - \frac{d}{c} \right)$$

$$= \left(\frac{ad - bc}{bd} \right) \left(\frac{bc - ad}{ac} \right)$$

$$= - \left(\frac{ad - bc}{bd} \right) \left(\frac{ad - bc}{ac} \right)$$

$$= - \frac{(ad - bc)^2}{abcd}$$

≤ 0 as required. $as (ad - bc)^2 \geq 0$ (perfect square)
 $abcd > 0$ (given)

(c)

$$\int_1^4 \frac{3}{x + 4\sqrt{5-x}} dx$$

$$= \int_{u=2}^{u=1} \frac{3}{(5-u^2) + 4\sqrt{u^2}} \cdot -2u du$$

$$= \int_1^2 \frac{6u}{5-u^2+4u} du$$

$$u^2 = 5-x$$

$$x = 5-u^2$$

$$dx = -2u du$$

$$x=1 \quad u^2=4 \therefore u=2 \quad (u>0)$$

$$x=4 \quad u^2=1 \therefore u=1$$

$$= \int_1^2 \frac{6u}{(5-u)(u+1)} du$$

$$\frac{6u}{(5-u)(u+1)} = \frac{A}{5-u} + \frac{B}{u+1}$$

$$= \int_1^2 \left(\frac{5}{5-u} - \frac{1}{u+1} \right) du$$

$$6u = (u+1)A + (5-u)B$$

$$\text{let } u=5 \quad 6A=30 \therefore A=5$$

$$\text{let } u=-1 \quad 6B=-6 \therefore B=-1$$

$$= \left[-5 \ln(5-u) - \ln(u+1) \right]_1^2$$

$$= \left[5 \ln(5-u) + \ln(u+1) \right]_2^1$$

$$= 5 \ln 4 + \ln 2 - (5 \ln 3 + \ln 3)$$

$$= \ln \left(\frac{4^5 \times 2}{3^5 \times 3} \right)$$

$$= \ln \left(\frac{2048}{729} \right) \text{ as required.}$$

$$(d) \quad z = \cos \theta - i \sin \theta$$

$$= \cos(-\theta) + i \sin(-\theta)$$

$$= e^{i(-\theta)}$$

$$z^n + \frac{1}{z^n} = z^n + z^{-n}$$

$$= e^{i(-n\theta)} + e^{i(n\theta)} \quad (\text{de Moivre's})$$

Q13 d (i) continued

$$\begin{aligned}
 z^n + \frac{1}{z^n} &= \cos(-n\theta) + i\sin(-n\theta) + \cos(n\theta) + i\sin(n\theta) \\
 &= \cos n\theta - i\sin n\theta + \cos n\theta + i\sin n\theta \quad \left(\begin{array}{l} \text{sin odd} \\ \text{cos even} \end{array} \right) \\
 &= 2\cos n\theta \text{ as required.}
 \end{aligned}$$

$$\begin{aligned}
 \text{(i)} \quad \left(z + \frac{1}{z}\right)^5 &= z^5 + 5z^4 \times \frac{1}{z} + 10z^3 \times \frac{1}{z^2} + 10z^2 \times \frac{1}{z^3} + 5z \times \frac{1}{z^4} + \frac{1}{z^5} \\
 &= z^5 + 5z^3 + 10z + \frac{10}{z} + \frac{5}{z^3} + \frac{1}{z^5} \\
 &= \left(z^5 + \frac{1}{z^5}\right) + 5\left(z^3 + \frac{1}{z^3}\right) + 10\left(z + \frac{1}{z}\right) \\
 &= 2\cos 5\theta + 10\cos 3\theta + 20\cos \theta \\
 \therefore A &= 2 \quad B = 10 \text{ and } C = 20
 \end{aligned}$$

$$\text{(ii)} \quad \left(z^5 + \frac{1}{z^5}\right)^5 = (2\cos \theta)^5 = 32\cos^5 \theta$$

so using Part (i)

$$32\cos^5 \theta = 2\cos 5\theta + 10\cos 3\theta + 20\cos \theta$$

$$\text{so } \cos^5 \theta = \frac{\cos 5\theta + 5\cos 3\theta + 10\cos \theta}{16}$$

$$\int_0^{\pi/2} \cos^5 \theta \, d\theta = \frac{1}{16} \int_0^{\pi/2} \cos 5\theta + 5\cos 3\theta + 10\cos \theta \, d\theta$$

$$\begin{aligned}
 &= \frac{1}{16} \left[\frac{\sin 5\theta}{5} + \frac{5 \sin 3\theta}{3} + 10 \sin \theta \right]^{11/2} \\
 &= \frac{1}{16} \left[\left(\frac{\sin 5\pi/2}{5} + \frac{5 \sin 3\pi/2}{3} + 10 \sin \pi/2 \right) - (0) \right] \\
 &= \frac{1}{16} \left[\frac{1}{5} - \frac{5}{3} + 10 \right] \\
 &= \frac{8}{15}
 \end{aligned}$$

Question 14

(a)(i) $a \pm v\sqrt{4-v^2}$

(ii) $\ddot{x}$ always +ve

$$x \frac{dv}{dx} = x\sqrt{4-v^2}$$

as $\dot{x}$ initially > 0 so $v\sqrt{4-v^2} > 0$

$$\int_1^v \frac{dv}{\sqrt{4-v^2}} = \int_0^{\pi/6} x \, dx$$

as $t \rightarrow \infty$ $v \rightarrow 2$ $x \rightarrow \infty$

The particles velocity increases
and approaches 2 m/s

$$\left[\arcsin\left(\frac{v}{2}\right) \right]_1^v = \left[x \right]_0^{\pi/6}$$

$$\arcsin\left(\frac{v}{2}\right) - \arcsin\left(\frac{1}{2}\right) = \pi/6$$

$$\arcsin\left(\frac{v}{2}\right) = \pi/3$$

$$\frac{v}{2} = \frac{\sqrt{3}}{2}$$

$$\therefore v = \sqrt{3} \text{ m/s}$$

Q14 continued.

(b)

$$V = \pi \int_0^{\sqrt{3}} (\sqrt{2x \arctan x})^2 dx$$

$$= \pi \int_0^{\sqrt{3}} 2x \arctan x dx$$

$$u = \arctan x$$

$$\frac{du}{dx} = \frac{1}{1+x^2}$$

$$\frac{dv}{dx} = 2x$$

$$v = x^2$$

$$= \pi \left[uv \right]_0^{\sqrt{3}} - \int_0^{\sqrt{3}} \frac{du}{dx} v dx$$

$$= \pi \left[\left[x^2 \arctan x \right]_0^{\sqrt{3}} - \int_0^{\sqrt{3}} \frac{x^2}{1+x^2} dx \right]$$

$$= \pi \left[\left[(\sqrt{3})^2 \cdot \frac{\pi}{3} - 0 \right] - \int_0^{\sqrt{3}} \left(\frac{x^2+1}{1+x^2} - \frac{1}{1+x^2} \right) dx \right]$$

$$= \pi \left[\pi - \int_0^{\sqrt{3}} \left(1 - \frac{1}{1+x^2} \right) dx \right]$$

$$= \pi^2 - \pi \left[x - \arctan x \right]_0^{\sqrt{3}}$$

$$= \pi^2 - \pi \left[(\sqrt{3} - \frac{\pi}{3}) - (0-0) \right]$$

$$= \pi^2 - \sqrt{3}\pi + \frac{\pi^2}{3}$$

$$= \frac{4\pi^2}{3} - \sqrt{3}\pi$$

Q14 (c)

$$\text{assume } 4x^2 - y^2 = 1 \quad x, y \in \mathbb{Z}$$

$$(2x)^2 - y^2 = 1$$

$$(2x+y)(2x-y) = 1$$

$$\text{as } x, y \in \mathbb{Z} \quad (2x+y) \in \mathbb{Z} \quad \text{and} \quad (2x-y) \in \mathbb{Z}$$

$\therefore$ only possible solutions are.

Case 1

$$\begin{aligned} (2x+y) &= 1 \quad \text{--- ①} \\ (2x-y) &= 1 \quad \text{--- ②} \end{aligned} \quad \text{or}$$

Case 2

$$\begin{aligned} (2x+y) &= -1 \quad \text{--- ③} \\ (2x-y) &= -1 \quad \text{--- ④} \end{aligned}$$

$$\text{①} + \text{②}$$

$$4x = 2 \quad \therefore x = \frac{1}{2}$$

which contradicts
assumption.

$$\text{③} + \text{④}$$

$$4x = -2$$

$$\therefore x = -\frac{1}{2}$$

which also contradicts
assumption.

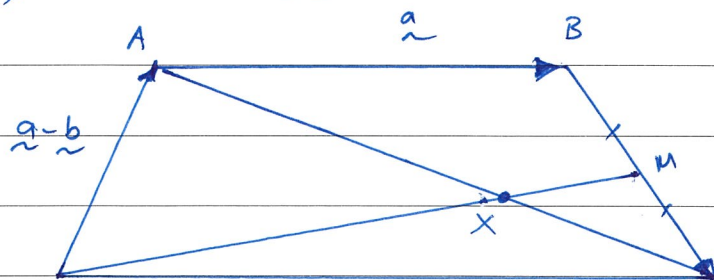
since both cases lead to non-integer values of x

original assumption must be wrong.

Therefore by contradiction $4x^2 - y^2 = 1$ has no
integer solutions.

Q14 continued.

(d)



$$OX:OM = n:1$$

$$b = \vec{BC}$$

$$a = \vec{AB}$$

$$a - b = \vec{OA}$$

$$\begin{aligned} (i) \quad \vec{OC} &= \vec{OA} + \vec{OB} + \vec{BC} \\ &= a - b + a + b \\ &= 2a \end{aligned}$$

$$(ii) \quad \vec{OX} = \frac{n}{n+1} \vec{OM}$$

$$= \frac{n}{n+1} (\vec{OB} + \vec{BM})$$

$$= \frac{n}{n+1} (a - b + a + \frac{1}{2}b)$$

$$= \frac{n}{n+1} (2a - \frac{1}{2}b)$$

$$\vec{OX} = \frac{2n}{n+1} a - \frac{n}{2(n+1)} b$$

$$\vec{AX} = \vec{AO} + \vec{OX}$$

$$= \underline{b} - \underline{a} + \frac{2n}{n+1} \underline{a} - \frac{n}{2(n+1)} \underline{b}$$

$$= \frac{2n - (n+1)}{n+1} \underline{a} + \frac{2(n+1) - n}{2(n+1)} \underline{b}$$

$$\vec{AX} = \frac{n+1}{n+1} \underline{a} + \frac{n+2}{2(n+1)} \underline{b} \quad \text{as required.}$$

$$(iii) \vec{AC} = \underline{a} + \underline{b}$$

X also lies on line $\vec{AC}$

$$\vec{AX} = \lambda \vec{AC}$$

$$\vec{AX} = \lambda (\underline{a} + \underline{b})$$

equating coefficients for $\underline{a}$ and $\underline{b}$ components using Part ii,

$$\left(\frac{n+1}{n+1} \right) = \lambda \quad \text{and} \quad \frac{n+2}{2(n+1)} = \lambda$$

i.e

$$\frac{(n+1)}{n+1} = \frac{n+2}{2(n+1)} \quad (n > 0)$$

$$2(n+1) = n+2$$

$$2n+2 = n+2$$

$$\therefore \underline{n = 4}$$

Question 15

(a) $Q \Rightarrow P$ Contrapositive $\neg P \Rightarrow \neg Q$

$$Q = \neg(A \Rightarrow C)$$

$$P = (\neg(A \Rightarrow B) \text{ OR } \neg(B \Rightarrow C))$$

$$\neg Q = \neg \neg(A \Rightarrow C) = A \Rightarrow C$$

$$\neg P = \neg (\neg(A \Rightarrow B) \text{ OR } \neg(B \Rightarrow C))$$

$$= ((A \Rightarrow B) \text{ AND } (B \Rightarrow C))$$

$\therefore$ Contrapositive is

$$(A \Rightarrow B) \text{ AND } (B \Rightarrow C) \Rightarrow (A \Rightarrow C)$$

also accept:

$$\neg (\neg(A \Rightarrow B) \text{ OR } \neg(B \Rightarrow C)) \Rightarrow (A \Rightarrow C)$$

(b) (i)

$$I_n = \int_0^1 x(1-x)^n dx$$

$$\frac{dv}{dx} = (1-x)^n$$

$$v = -\frac{(1-x)^{n+1}}{n+1}$$

$$= [uv]_0^1 - \int_0^1 v \frac{du}{dx} dx$$

$$u = x$$

$$\frac{du}{dx} = 1$$

$$= \left[\frac{-(1-x)^{n+1}}{n+1} \cdot x \right]_0^1 - \int_0^1 -\frac{(1-x)^{n+1}}{n+1} dx$$

$$= [0 - 0] + \int \frac{(1-x)^{n+1}}{n+1} dx$$

$$= \left[-\frac{(1-x)^{n+2}}{(n+1)(n+2)} \right]_0^1 = \left[\frac{(1-x)^{n+2}}{(n+1)(n+2)} \right]_1^0$$

$$= \frac{(1-0)^{n+2}}{(n+1)(n+2)} - 0$$

$$= \frac{1}{(n+2)(n+1)} \text{ as required.}$$

$$(ii) \quad S_n = \sum_{k=1}^n \frac{1}{(k+1)(k+2)}$$

$$\frac{1}{(n+1)(n+2)} = \frac{A}{n+1} + \frac{B}{n+2}$$

$$1 = (n+2)A + (n+1)B$$

$$\text{let } n = -1$$

$$1 = B$$

$$\text{let } n = -2$$

$$-B = 1 \therefore B = -1$$

$$= \sum_{k=1}^n \left(\frac{1}{k+1} - \frac{1}{k+2} \right)$$

$$= \left(\frac{1}{2} - \frac{1}{3} \right) + \left(\frac{1}{3} - \frac{1}{4} \right) + \dots + \left(\frac{1}{n+1} - \frac{1}{n+2} \right)$$

$$= \frac{1}{2} - \frac{1}{n+2}$$

$$= \frac{n+2-2}{2(n+2)}$$

$$S_n = \frac{n}{2(n+2)}$$

(P15) continued.

(c)

$$\text{RTP } \tan \theta + 2 \tan 2\theta + \dots + 2^{n-1} \tan(2^{n-1} \theta) = \cot \theta - 2^n \cot(2^n \theta)$$

for $n=1$

$$\text{LHS} = \tan \theta$$

$$\text{RHS} = \cot \theta - 2 \cot(2\theta)$$

$$= \cot \theta - 2 \left(\frac{1 - \tan^2 \theta}{2 \tan \theta} \right)$$

$$= \cot \theta - \frac{1}{\tan \theta} + \frac{\tan^2 \theta}{\tan \theta}$$

$$= \cot \theta - \tan \theta + \tan \theta$$

$$= \tan \theta$$

$$= \text{RHS}$$

$\therefore$ true for $n=1$

assume true for $n=k$.

$$\text{i.e. } \tan \theta + 2 \tan 2\theta + \dots + 2^{k-1} \tan(2^{k-1} \theta) = \cot \theta - 2^k \cot(2^k \theta)$$

RTP for $n=k+1$

i.e. RTP

$$\tan \theta + 2 \tan 2\theta + \dots + 2^{k-1} \tan(2^{k-1} \theta) + 2^k \tan(2^k \theta) = \cot \theta - 2^k \cot(2^k \theta) + 2^k \tan(2^k \theta)$$

for $n=k+1$

$$\text{LHS} = \tan \theta + 2 \tan 2\theta + \dots + 2^{k-1} \tan(2^{k-1} \theta) + 2^k \tan(2^k \theta)$$

$$= \cot \theta - 2^k \cot(2^k \theta) + 2^k \tan(2^k \theta) \text{ from assumption}$$

$$\begin{aligned}
\text{LHS} &= \cot \theta - 2^k (\cot(2^k \theta) - \tan(2^k \theta)) \\
&= \cot \theta - 2^k \left(\frac{\cos(2^k \theta)}{\sin(2^k \theta)} - \frac{\sin(2^k \theta)}{\cos(2^k \theta)} \right) \\
&= \cot \theta - \frac{2^{k+1}}{2} \left(\frac{\cos(2^k \theta) \cdot \cos(2^k \theta) - \sin(2^k \theta) \cdot \sin(2^k \theta)}{\sin(2^k \theta) \cos(2^k \theta)} \right) \\
&= \cot \theta - 2^{k+1} \left(\frac{\cos(2^{k+1} \theta) - \sin(2^k \theta)}{2 \sin(2^k \theta) \cos(2^k \theta)} \right) \\
&= \cot \theta - 2^{k+1} \left(\frac{\cos(2^{k+1} \theta)}{\sin(2^{k+1} \theta)} \right) \\
&= \cot \theta - 2^{k+1} \cot(2^{k+1} \theta) \\
&= \text{RHS}
\end{aligned}$$

$\therefore$ true for $n=1$ if true for $n=k$.

but true for $n=1$ so true for $n=2, 3, \dots$

$\therefore$ by the process of mathematical induction.

true $\forall n \in \mathbb{Z}^+$

alternate inductive step.

Note after assumption we can use the result from step 1 but we need to structure carefully.

$$\begin{aligned}
\text{LHS} &= \cot \theta - 2^k \cot(2^k \theta) + 2^k \tan(2^k \theta) \\
&= \cot \theta - 2^k (\cot(2^k \theta) - \tan(2^k \theta))
\end{aligned}$$

now let $\alpha = 2^k \theta$

$$\text{LHS} = \cot \theta - 2^k (\cot \alpha - \tan \alpha)$$

From $n=1$ step.

$$\tan \theta = \cot \theta - 2 \cot 2\theta$$

$$\therefore \cot \alpha - \tan \alpha = 2 \cot(2\alpha)$$

$$= \cot \theta - 2^k (2 \cot 2\alpha)$$

$$= \cot \theta - 2 \cdot 2^k \cot (2 \cdot 2^k \theta) \quad (\text{replacing } \alpha = 2^{k+1} \theta)$$

$$= \cot \theta - 2^{k+1} \cot 2^{k+1} \theta$$

$$= \text{RHS}$$

etc

(d) Fundamental Theorem of Calculus $\frac{d}{dx} \int_a^x f(t) dt = f(x)$

let $g(x) = \int_1^x \ln(t) dt$ $f(t) = \ln(t)$

$$\frac{d}{dx} \int_1^{x^2} \ln t dt = \frac{d}{dx} g(x^2)$$

$$= g'(x^2) \cdot \frac{d}{dx} x^2 \quad (\text{chain rule})$$

$$= 2x g'(x^2)$$

$$= 2x f(x^2) \quad (\text{fundamental theorem of calculus})$$

$$= 2x \ln(x^2)$$

$$= 2 \cdot 2x \ln(x)$$

$$= 4x \ln x \quad \text{as required.}$$

(P15)(e) $F_k \leftarrow \boxed{m_1} - \boxed{m_2} \rightarrow m_2 g \rightarrow \text{direction of travel}$

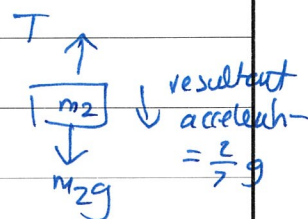
$$\Sigma F = m_2 g - F_k$$

$$= 6g - \frac{1}{4} \times 8 \times g = 4g$$

$F = ma$ where m is the mass of whole system ($M = 6+8 = 14$)

$\therefore 4g = 14a$ i.e. $a = \frac{4g}{14}$ $a = \frac{2}{7}g$

For tension look at part of system



$$\Sigma F = (m_2 g - T) = 6g - T \quad \text{--- (1)}$$

also $F = ma$.

$$F = 6 \times \frac{2}{7}g = \frac{12g}{7} \quad \text{--- (2)}$$

(1) = (2)

$$6g - T = \frac{12g}{7} \quad \text{so } T = 6g - \frac{12g}{7}$$

$$T = \frac{42 - 12}{7}g \text{ N}$$

$$T = \frac{30}{7}g \text{ N}$$

Question (16)

(a)(i) $2n$ and $2n+2$ $n \in \mathbb{Z}^+$ are 2 positive consecutive integers.

$$2n(2n+1) = 4 \cdot n \cdot (n+1)$$

n and $n+1$ are consecutive integers

$$= 4 \cdot 2k \quad k \in \mathbb{Z}^+$$

if n even then $n+1$ odd
if n odd then $n+1$ even.

$$= 8k$$

$$\therefore n(n+1) = 2k \quad k \in \mathbb{Z}^+$$

which is divisible by 8

(p6)(ii) $(p^2-1) = (p+1)(p-1)$

p is Prime and $> 3 \therefore p$ is odd

$\therefore (p+1)$ and $(p-1)$ are consecutive even integers.

$\therefore$ from Part(i) $(p+1)(p-1) = 8m \quad m \in \mathbb{Z}^+$

now consider the 3 consecutive integers $(p-1), p, (p+1)$

one of the 3 must be divisible by 3

however p is Prime $> 3 \therefore$ not divisible by 3

i.e either $(p-1)$ divisible by 3 or $(p+1)$ divisible by 3

$\therefore (p-1)(p+1) = 3p \quad p \in \mathbb{Z}^+$

as 8 and 3 are co prime.

$(p-1)(p+1) = 3 \times 8 \times s \quad s \in \mathbb{Z}^+$

$\therefore p^2-1 = 24s$

i.e p^2-1 is divisible by 24 if p is a Prime > 3

(b)(i) $\sum F = F_r + F_{\text{gravity}}$ $\uparrow_{+ve} \quad \uparrow_{F_r} \quad \downarrow_{F_{\text{gravity}}}$

$$= \underbrace{\frac{mg}{\gamma^3} (1+v^2)}_{F_r} - \underbrace{mg}_{F_g}$$

$$F_{\text{total}} = mg \left(\frac{1+v^2}{\gamma^3} - 1-v^2 \right) =$$

$$ma = mg \left(\frac{1+v^2}{\gamma^3} - (1+v^2) \right)$$

$$v \frac{dv}{dc} = g(1+v^2) \left(\frac{1}{\gamma^3} - 1 \right)$$

$$\frac{v}{1+v^2} \frac{dv}{dc} = -g \left(1 - \frac{1}{\gamma^3} \right) \text{ as required.}$$

Q16(b)(ii)

$$\frac{v}{1+v^2} \frac{dv}{dx} = -g \left(1 - \frac{1}{x^3}\right)$$

initially $V = u$

$$v_{\text{max height}} = 0$$

initially $x = 1$

$$x_{\text{max height}} = H$$

$$\int_u^0 \frac{v}{1+v^2} dv = - \int_1^H g \left(1 - \frac{1}{x^3}\right) dx$$

 u is positive somax height when $V = 0$

what happens after

this is beyond

the scope of

the question

$$\frac{1}{2} \int_u^0 \frac{2v}{1+v^2} dv = \int_1^H \frac{g}{x^3} - g dx$$

$$\left[\frac{1}{2} \ln(1+v^2) \right]_u^0 = \int_1^H g x^{-3} - g dx$$

$$\frac{1}{2} \ln(1) - \frac{1}{2} \ln(1+u^2) = \left[-\frac{g x^{-2}}{2} - g x \right]_1^H$$

$$-\frac{1}{2} \ln(1+u^2) = \left[-\frac{g x^{-2}}{2} - g x \right]_1^H$$

$$\ln(1+u^2) = \left[g x^{-2} + 2g x \right]_1^H$$

$$= \left[\frac{g}{H^2} - g + 2gH - 2g \right]$$

$$= 2g(H-1) + g\left(\frac{1}{H^2} - 1\right)$$

$$\ln(1+u^2) = 2g(H-1) + g\left(\frac{1-H^2}{H^2}\right)$$

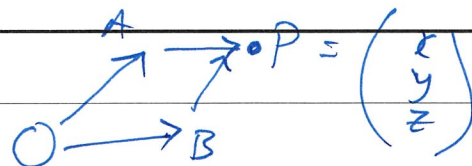
$$\ln(1+u^2) = 2g(H-1) - g\left(\frac{H^2-1}{H^2}\right)$$

as required.

Q16cc)

$$\vec{OA} = \begin{pmatrix} 6 \\ 4 \\ -2 \end{pmatrix}$$

$$\vec{OB} = \begin{pmatrix} 4 \\ 8 \\ 2 \end{pmatrix}$$



$$\vec{AP} = \begin{pmatrix} x-6 \\ y-4 \\ z+2 \end{pmatrix}$$

$$\vec{BP} = \begin{pmatrix} x-4 \\ y-8 \\ z-2 \end{pmatrix}$$

$$\vec{AP} \cdot \vec{BP} = 0$$

$$(x-6)(x-4) + (y-4)(y-8) + (z+2)(z-2) = 0$$

$$x^2 - 10x + 24 + y^2 - 12y + 32 + z^2 - 4 = 0$$

$$x^2 - 10x + y^2 - 12y + z^2 + 52 = 0$$

$$(x-5)^2 - 25 + (y-6)^2 - 36 + z^2 + 52 = 0$$

$$(x-5)^2 + (y-6)^2 + z^2$$

This is a circle radius 3 centred on $\begin{pmatrix} 5 \\ 6 \\ 0 \end{pmatrix}$

$$\therefore \left| \underline{v} - \begin{pmatrix} 5 \\ 6 \\ 0 \end{pmatrix} \right| = 3 \text{ is the vector equation of } S_1$$

alternately: we observe that in 2D the angle subtended by 2 points on a diameter of a circle equals 90° (Thales' theorem)
 $\therefore$ in 3D this will be a sphere, centre is midpoint of AB
 and radius equals $\frac{1}{2}$ length of segment AB

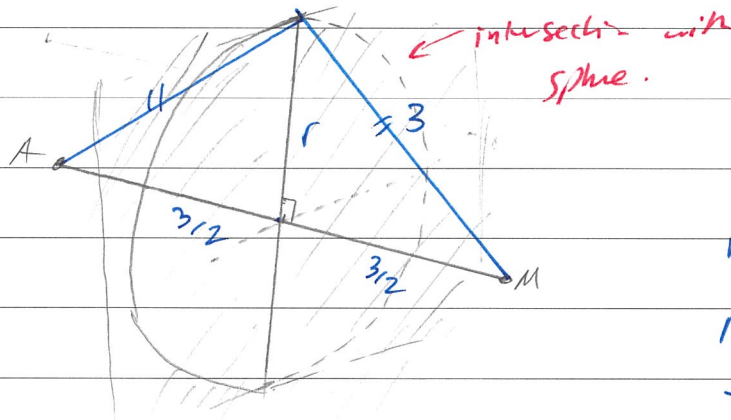
$$\vec{BA} = \begin{pmatrix} 6-4 \\ 4-8 \\ 2-2 \end{pmatrix} = \begin{pmatrix} 2 \\ -4 \\ 0 \end{pmatrix}$$

$$\frac{|\vec{BA}|}{2} = \frac{\sqrt{2^2 + (-4)^2 + 0^2}}{2} = \frac{\sqrt{36}}{2} = \underline{\underline{3}}$$

$$\text{and } M = \begin{pmatrix} \frac{6+4}{2} \\ \frac{4+8}{2} \\ \frac{-2+2}{2} \end{pmatrix} = \begin{pmatrix} 5 \\ 6 \\ 0 \end{pmatrix} \therefore S_1 \in \left| \underline{v} - \begin{pmatrix} 5 \\ 6 \\ 0 \end{pmatrix} \right| = 3$$

Q16) (c) (ii) $|\vec{AN}| = |\vec{MN}|$ in 2D N would be the line.

that was the perpendicular bisector of AM in 3D this extends up and down to a plane whose intersection with sphere from part (c) gives us our circle.



$$r^2 = 3^2 - \left(\frac{3}{2}\right)^2$$

$$= 9 - \frac{9}{4} = \frac{27}{4}$$

$$r = \frac{\sqrt{27}}{2}$$

$$r = \frac{3\sqrt{3}}{2}$$

(d)

$$\int_a^b f(x) dx$$

let $x = a + b - u$

$dx = -du$

$$= \int_{u=b}^a f(a+b-u) \cdot -du$$

$x=b \rightarrow u=a$
 $x=a \rightarrow u=b$

$$= - \int_b^a f(a+b-u) du$$

$$= \int_a^b f(a+b-u) du$$

$$= \int_a^b f(a+b-x) dx$$

as definite integral independent of
 the variable used
 "dummy variable"

Q16(d)

$$(ii) I = \int_{\pi/6}^{\pi/3} \frac{1}{1 + \sqrt{\tan x}} dx$$

[alternately split into

$$\sqrt{\frac{\sin x}{\cos x}} \text{ and } x \sqrt{\frac{\cos x}{\sin x}}]$$

$$= \int_{\pi/6}^{\pi/3} \frac{1}{1 + \sqrt{\tan(\pi/2 - x)}} dx$$

from part (i) $\left(\frac{a+b-x}{\pi/3 + \pi/6 = \pi/2} \right)$

$$= \int_{\pi/6}^{\pi/3} \frac{1}{1 + \sqrt{\cot x}} dx$$

$$= \int_{\pi/6}^{\pi/3} \frac{1}{1 + \sqrt{\frac{1}{\tan x}}} dx = \int_{\pi/6}^{\pi/3} \frac{\sqrt{\tan x}}{\sqrt{\tan x} + 1} dx \quad \left(\times \frac{\sqrt{\tan x}}{\sqrt{\tan x}} \right)$$

$$I = \int_{\pi/6}^{\pi/3} \frac{\sqrt{\tan x} + 1 - 1}{\sqrt{\tan x} + 1} dx$$

$$I = \int_{\pi/6}^{\pi/3} 1 - \frac{1}{\sqrt{\tan x} + 1} dx = \int_{\pi/6}^{\pi/3} 1 dx - I$$

$$2I = \int_{\pi/6}^{\pi/3} 1 dx = \pi/3 - \pi/6$$

$$\therefore I = \pi/12$$

The end